

Joel Bhattarai

joelbh@umich.edu | [248-974-4375](tel:248-974-4375) | [linkedin.com/in/joel-bhattarai](https://www.linkedin.com/in/joel-bhattarai) | East Palo Alto, California

EDUCATION

University of Michigan

Bachelor of Science in Engineering in Mechanical Engineering

Ann Arbor, MI

May 2026

EXPERIENCE

Automation Engineer

3D Infotech

Irvine, CA

August 2026 – Present

- Completed training in PolyWorks metrology and FANUC and Universal Robots scripting and operation.
- Implemented fully in-line and semi-in-line systems from design to integration, testing, and deployment.
- Gathered and analyzed metrology data for a variety of different geometries and applications.

Principal Investigator

NASA L'SPACE Program - Proposal Writing and Evaluation Experience

May 2025 – September 2025

- Independently generated a new technology within the areas of Autonomy, Robotics, and AI that was selected for a full proposal by a NASA panel.
- Led a group of ten interdisciplinary students to a full proposal and an NTR (New Technology Report).
- Advanced proposed technology from concept to final proposal.
- Served as the primary reviewer on a NASA proposal review board.

Lead Systems Engineer

NASA L'SPACE Program - Mission Concept Academy

May 2024 – September 2024

- Managed a top ten performing team of engineering students to a successful PDR.
- Designed a rover to accomplish practical missions on the surface of the moon.
- Independently developed system interface and wiring diagrams while managing overall rover metrics such as weight, power, and size.
- Completed skills modules on Heat Transfer, Siemens NX, Teaming, Engineering Requirements, Project Management, Systems Topics, Risk Management.

PROJECTS

Inverted Pendulum Stabilization

University of Michigan Automotive Laboratory

Ann Arbor, MI

November 2025

- Modeled physical behavior of inverted pendulum setup in state space for control design.
- Numerically determined multiple gain values using pole placement for a full-state feedback controller.
- Leveraged Simulink to apply solution to physical hardware on Quanser Qube Servo 3 and evaluated results based on real-world performance.
- Reduced settling time to less than 2 seconds with a maximum control input of 10 V.

SKILLS

Software & Simulation: PolyWorks, SolidWorks, Siemens NX, HyperMesh, HyperWorks, OptiStruct, ANSYS, LaTeX, Microsoft Azure AI Fundamentals Certified

Controls & Computing: MATLAB, Simulink, LabVIEW, Arduino, C/C++

Manufacturing & Lab: Machining (Mill & Lathe), Metalworking, Waterjetting, Geometric Dimensioning & Tolerancing (GD&T), Oscilloscopes, Multimeters, Strain Gauges, Thermocouples

COURSES

Relevant Coursework

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|----------------------------------|------------------------------|---------------------------|
| • Solid Mechanics and Statics | • Fluid Mechanics | • Automatic Control |
| • Thermodynamics | • Heat Transfer | • Energy Infrastructure |
| • Dynamics and Vibrations | • Control of Dynamic Systems | • Modern Energy Materials |
| • Design and Manufacturing I-III | • Behavior of Materials | |
| • Finite Elements | • Electric Circuits | |